INCH-POUND
A-A-60054
April 11, 2023

COMMERCIAL ITEM DESCRIPTION

SOCKS, INTENSE COLD WEATHER

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description covers intense cold weather socks intended for wear by personnel of the Department of Defense.

2. CLASSIFICATION. The socks are available in one type in the following sizes:

2.1 Size. Socks are sized according to the wearer's normal shoe size, as shown in table I.

TABLE I. Sock sizes.

Sock size	Men's shoe size	Women's shoe size
Small (SM)	N/A	3 to 6.5
Medium (MD)	4 to 8.5	7 to 10
Large (LG)	9 to 12	10.5 to 14
Extra Large (XL)	12.5 to 16	N/A

3. SALIENT CHARACTERISTICS.

3.1 Description. The socks are over-the-calf length with sandwich terry cushioning comfort and graduated compression to promote blood flow from the foot with a non-slip top cuff opening. The foot of each sock includes contouring for the sock to stay in place without bunching, sandwich terry cushion sole, a gore heel, nylon reinforcement at the heel and toe, arch support, instep ventilation. The sock has a seamless toe closure or a knitted closure for comfort and durability.

3.2 Basic materials. The finished socks shall contain wool at a minimum grade not coarser than US 60s to 62s (25 microns or finer), nylon, and spandex, with no other materials included (see 3.6).

3.3 Construction. The finished socks shall be knitted with the sock contour construction features listed in table II and shall meet the sock weight specified in 3.7. Socks knitted on a 108-needle machine have been shown to meet these requirements. Other types of knitting machines may be used as long as they meet all requirements of this commercial item description (see 7.11).

Beneficial comments, recommendations, additions, deletions, clarifications, and any other data that may improve this document should be sent to: Marine Corps Systems Command, 2200 Lester Street, Quantico, VA 22134 ATTN: SEAL-SE-STDS or emailed to USMC_STDZ@usmc.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

FSC 8415

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TABLE II. Contour sock construction.

Sock region	Construction
Top cuff opening	1 x 1 rib with elastic at a nominal 26 courses per inch in double no-ravel cuff. The size marking (see 3.9) is knit onto the exterior cuff surface in a light contrasting color yarn for easy visibility.
Leg	Sandwich terry knit with nominal 22 courses per inch in 3 x 1 rib with stretch yarns plaited in the knit for graduated compression.
Foot cushion sole	Sandwich full terry knit with nominal 34 courses per inch.
Heel	Gore heel with nylon reinforcement. Y gore is preferred.
Arch support	100 percent elastic yarn feeds, which circle the foot in the middle section.
Instep	A mesh knit included for ventilation.
Toe	Full terry, nylon reinforced toe, formed with automatic toe closure (no raised seam).

3.4 Color. When tested in accordance with 3.5, the color shall match Black 477, AMS-STD-595 color chip 37029 (see 7.4.2). Alternative colors may be specified in the contract or purchase order (see 7.9).

3.5 Visual shade matching. The sock shall match the standard shade sample (see 3.4) when viewed using AATCC EP 9, option C (see 7.4), with a primary light source simulating average daylight, CIE Illuminant D65, with a color temperature of 6500±200 Kelvin (K) and illumination of 100±20 foot candles. Alternatively, the color and appearance of the sock shall match the standard sample with a primary light source simulating artificial daylight, CIE Illuminant D75, with a color temperature of 7500±200K and illumination of 100±20 foot candles in lieu of D65 (see 7.4.1). The sock shall also be a good match to the standard sample with a secondary source simulating incandescent lamplight, CIE Illuminant A, with a color temperature of 2856±200K.

3.6 Performance requirements. The finished socks shall conform to the requirements specified in table III when tested as specified.

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TABLE III. Sock performance requirements.

Characteristic	Requirement	Test method
Fiber analysis, percent 1/		
Wool (minimum) (see 3.2)	83	AATCC TM20A
Nylon (minimum)	14	
Spandex (maximum)	3	
Thickness, sole area, inch at 0.15 pounds per square inch (psi)	0.16 to 0.20	ASTM D1777
Bursting strength, sock heel, pounds force (lbf) (minimum)	75	ASTM D6797
Dimensional stability, percent (maximum)		
Foot length	6.0	AATCC TM150 1, IV, Aiii, 3 cycles 3.6.1
Calf length	6.0	
Foot width	7.0	
Elastic recovery, top cuff opening, percent after one hour (minimum)	85	3.6.2
Graduated compression, millimeters of mercury (mmHg), average		
Ankle	8 to 15	3.6.3
Calf	65 to 85 percent of the actual measured average ankle compression	3.6.3
Colorfastness, rating (minimum)		
Laundering, accelerated	3	AATCC TM61, 1A, 3 cycles AATCC EP1
Perspiration, acid/alkaline	4	AATCC TM15 AATCC EP1
Crocking, dry and wet	1.5	AATCC TM8 AATCC EP8

FOOTNOTE:

1/ Total overall fiber content may vary by ±3 percent from the labeled content as long as the final fiber content adds up to 100 percent.

3.6.1. Dimensional stability. Dimensional stability shall be calculated using finished sock dimensions for leg length, foot length, and foot width identified in figure 1, dimensions A, C, and D, when measured according to the methodology specified in 3.6.3. Socks shall be conditioned at 70±2 °F and 65±2 percent relative humidity in a relaxed condition before dimensional measurements are taken, initially and after three launderings (see 3.9.1).

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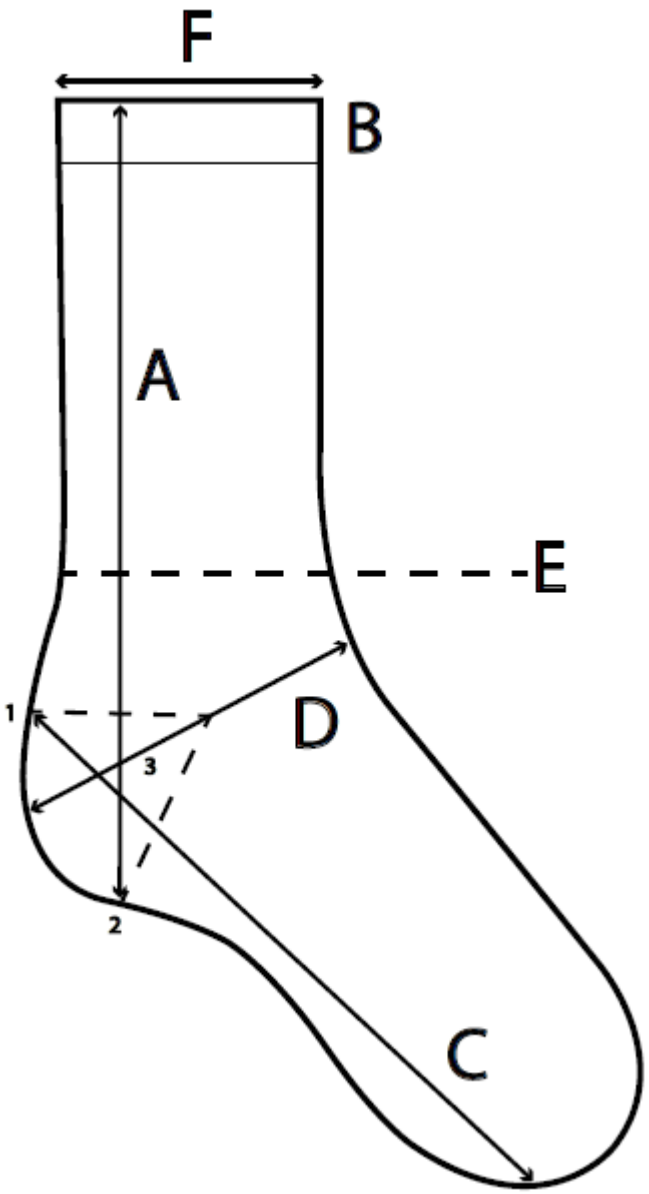


FIGURE 1. Sock with measurement lines.

3.6.2. Cuff stretch and elastic recovery. The sock cuff shall be tested intact on the sock in its original state to measure the cuff cross stretch and percent elastic recovery requirements specified in table IV. The cuff shall be tested in accordance with ASTM D2594/D2594M except that HIFOMACO LCS 4800 stretch device, or equal, shall be used in place of the hanger assembly. Alternately, cuff cross stretch may be measured with HIFOMACO scissor device (see 7.6), or equal. For the elastic recovery test, one cycle shall start at the unloaded opening and then be extended to 10-pound tension. At the fifth stretch cycle, the cuff shall be held extended for 10 seconds before unloading tension, then allowed to relax for 1 hour before measuring the cuff width again without tension.

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TABLE IV. Cuff measurements.

Property	Measurement recorded	Requirement
Cuff cross stretch, centimeter		
Initial	Initial cuff width recorded before stretch cycling	9.0±0.5
Extended	Extended width recorded at the fifth stretch cycle	10.25±0.5
Cuff elastic recovery, percent (minimum)	Initial cuff width Measured after 1 hour relaxation, centimeter	85 percent recovery 1/

FOOTNOTE:

1/ Calculation:

$$\frac{Initial\ width - (Stretched\ width\ at\ 5th\ cycle - Initial\ width)}{Initial\ width} \times 100 = \%$$

3.6.3 Graduated sock compression. The over-the-calf socks shall provide mild graduated compression measured in mmHg. Compression measurements shall comply with BS 661210, Graduated Compression Hosiery, Anti-Embolism Hosiery and Graduated Support Hosiery, using National Association of Hosiery Manufactures (NAHM) profile plates on Hosiery and Allied Trades Research Association (HATRA) type equipment (see 7.5). The compression measurements shall be read at the two circumference locations listed in table V. The compression shall be reported as an average and shall meet the requirement specified in table III, where the individual determinations shall not vary more than ±20 percent from the average.

TABLE V. Circumference measurements for compression measurements.

Compression measurement location	NAHM Profile Plate 11 (female size 9.5 to 10.5)	NAHM Profile Plate 13 (male size 12.0 to 12.5)
Ankle circumference (centimeters) 1/	23.15	24.93
Widest calf circumference (centimeters)	37.37	38.64

FOOTNOTE:

1/ Measurement taken at the horizontal plane measured 10 centimeters from the center of the heel gore line.

3.7 Fit and finished measurements. The socks shall conform to the NAHM board sizes and the finished measurements provided in table VI. The socks shall be unpackaged, placed flat, and relaxed while conditioning for 1 hour prior at 70±2 °F and 65±2 percent relative humidity, except as noted for the weight measurement below. The socks shall be measured in accordance with table VII measurement method and guidance.

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TABLE VI. Finished measurements.

Sock characteristic		Small	Medium	Large [2/]	Extra Large [2/]	Tolerance
Fit						
NAHM Board – male sock size		N/A	9 to 11	11 to 13	13 to 15	Pass/Fail
NAHM Board – female sock size		8.5 to 9.5	10 to 11	11.5 to 13	N/A	Pass/Fail
Finished measurement, inches						
A	Leg length	15	15-1/4	16	17	±3/4
B	Cuff length	1-1/4	1-1/4	1-1/4	1-1/4	±1/8
C	Foot length	8-3/4	9	10	11	±1/2
D	Foot width	3-7/8	3-7/8	3-7/8	3-7/8	±1/4
E	Reference for ankle compression measurement (see 3.6.3)	-	-	-	-	-
F	Cuff width	3-1/2	3-1/2	3-1/2	3-1/2	±1/2
Conditioned weight, ounces per dozen (oz/doz) [1/]		48	52.5	57	60	±3.6 oz/doz

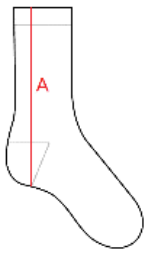
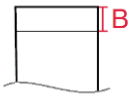
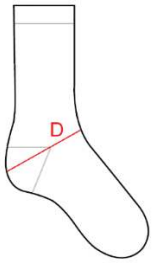
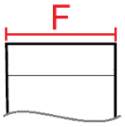
FOOTNOTE:

[1/] Scale weight after conditioning at least 4 hours at 70±2 °F and 65±2 percent relative humidity on a calibrated scale in ounce unit. Record the mass to one decimal place. If fewer than 12 pairs are available for assessment, the weight per dozen may be determined by multiplying the average weight per pair by 12.

[2/] The crossover of male size 11 in the Medium and Large sizes and male size 13 in the Large and Extra Large sizes is intentional because the calf and foot dimensions at those sizes could facilitate a better fit for some people with differing calf dimensions or foot shape/width.

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TABLE VII. Method of measuring socks.

Measurement	Description	Method guidance
Leg length (A)	With the ruler running parallel to the wale or ribbing direction in the leg, measure the leg from the top of the sock to the bottom of the heel at the transition point.	
Cuff top (B)	With a ruler running in the wale direction, measure from the top opening edge to the bottom cuff edge where the knit construction changes in the leg length.	 Note: Only cuff is shown
Foot length (C)	With the ruler running parallel to the wale direction in the foot, measure the foot length from the top of the heel at the transition point to the end of the toe.	
Foot width (D)	With the ruler placed on top of the gore line, which runs through the center of the heel, measure the foot width from the back of the heel at the transition point to the top of foot.	
Ankle compression test location (E)	The ankle test location is the horizontal plane measured 10 centimeters from the center of the heel gore line.	
Cuff width (F)	The cuff shall be flattened, and the width measured with a ruler in the course direction from the fold-to-fold edges.	 Note: Only cuff is shown

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3.8 Finish and pairing. The socks shall be scoured in finishing and boarded and matched in a pair where there is no greater than 1/2-inch difference in overall foot length and leg length between socks in a pair.

3.9 Label/marking. Sock sizes shall be Small, Medium, Large, and Extra Large and abbreviated to SM, MD, LG, and XL (see table I). Size Medium may also be abbreviated MED. The sock size shall be permanently knitted on the exterior of the cuff opening in a contrasting color that is legible at 3 feet. Two of the same size socks from the same production lot shall be packaged as a pair with a commercial type of label containing the contractor's name, style, size, fiber content breakdown, wash/care instructions, and UPC number. Laundering instructions shall state the socks can be machine washed in the normal cycle with hot water and tumble dried with hot setting.

3.9.1 Washing instructions. The socks shall maintain physical properties performance as specified in table III when washed in normal wash cycle with hot water (120±10 °F) and tumble dried hot (exhaust 155±10 °F). The socks are expected to withstand up to 50 launderings with no significant degradation in serviceability or durability.

3.10 End item dimensional examination. The socks shall be examined to verify conformance to the finished measurements in 3.7. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of socks. The sample unit shall be one pair of each size socks. The acceptance quality limits (AQL) shall be as specified in the contract or purchase order (see 7.9).

3.11 Toxicity test. When tested as specified herein, the finished socks shall not present a health hazard and shall show compatibility with prolonged direct skin contact. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used. When the toxicity requirement can be demonstrated with historical use data, toxicity testing may not be required on the finishing treatments used. If dermal toxicity testing is required (see 7.9), it should be conducted in accordance with 40 CFR § 798.2250, Dermal Toxicity, and 40 CFR § 798.4100, Dermal Sensitization, which are consistent with Organisation for Economic Co-operation and Development (OECD) Guidelines for the Testing of Chemicals, Section 4, Test Nos. 404, Acute Dermal Irritation/Corrosion, and 406, Skin Sensitisation (see 7.7).

3.12 Workmanship. The socks shall conform to the quality of product established by this commercial item description. The occurrence of defects shall not exceed the AQL (see 7.9).

4. REGULATORY REQUIREMENTS. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The finished socks provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance. The wool content and the graduated compression specified in table III may be addressed by a certificate of compliance.

5.2 Market acceptance. The finished socks shall have been sold to the commercial market or the government (see 7.8).

5.3 Visual examination. Examine the finished socks for the defects listed below as defined in ASTM D3990. Sampling for inspection shall be in accordance with ASQ/ANSI Z1.4.

5.3.1 Defects.

a. Any hole, runs, dropped stitch (including holes looping or seaming), cut, or tear
b. Color not as specified
c. Any part shaded, noticeable variation in size, or variation in cross stretch
d. Non-uniform in overall appearance
e. Not well shaped
f. Skew

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g. Slubby yarn

h. Stain

i. Shade bar or spot

j. Rancid or bad odor

k. Any burn or scorch

l. Skipped stitches

m. Bulky thread

n. Run-offs or broken thread

o. Any dimension or weight not as specified

p. Label missing, incorrect, misplaced, or illegible

q. Measurement of item not as specified

r. Bar code or UPC code omitted or not readable by a scanner (human readable interpretation), omitted or illegible

s. Bar code incorrect

t. Bar code not visible on packaged item

u. Bar code causes damage to the item

v. Not packaged in accordance with contract or purchase order

5.4 Acceptance criteria. Acceptance criteria shall be as specified in the contract or purchase order (see 7.9).

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order (see 7.9).

7. NOTES.

7.1 Intended use. The socks are intended for wear with an insulated intense cold weather boot by male and female military personnel of the Department of Defense operating in intense cold weather environments of -20 to +20 °F.

7.2 Source of documents.

7.2.1 AATCC test methods. AATCC test methods are available from www.aatcc.org or AATCC, P.O. Box 12215, Research Triangle Park, NC 27709.

7.2.2 ASQ standards. ASQ standards are available from www.asq.org or American Society for Quality, 600 North Plankinton Ave., Milwaukee, WI 53203.

7.2.3 ASTM International standards. ASTM standards are available from www.astm.org or ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428.

7.2.4 BSI standards. BSI standards are available from www.bsigroup.com or BSI Group, 12950 Worldgate Drive, Suite 800, Herndon, VA 20170.

7.2.5 Code of Federal Regulations (CFR). The CFR is available from www.ecfr.gov or the Superintendent of Documents, U.S. Government Publishing Office, Washington, DC 20401.

7.2.6 NAHM sizing standard, sizing boards, and compression profile plates. The NAHM sizing standard is available from www.legsource.com. NAHM sock sizing boards and compression profile plates are available from www.manufacturingsolutionscenter.org or Manufacturing Solutions Center, 301 Conover Station SE, Conover, NC 28613.

7.2.7 OECD guidelines. OECD guidelines are available from www.oecd-ilibrary.org or OECD Washington Center, 1776 I St NW, Suite 450, Washington, DC 20006.

7.2.8 SAE standards. AMS-STD-595 is available from www.sae.org or SAE World Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096.

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7.3 Standard samples. For access to standard samples, address the contracting activity issuing the invitation for bids or request for proposal.

7.4 Visual shade matching. In 2017, option A of AATCC EP 9 was changed to option C. NOTE: In case of confusion, the viewing geometry should be, "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35±5-degree angle relative to the horizontal. The observer will view the specimens at a 90-degree angle, relative to the plane of the specimens."

7.4.1 D65 illuminant. In 2022, the U.S. Military (all services) accepted D65 as the new preferred light source for visual shade matching.

7.4.2 Shade criticality. Some items may be deemed "non-shade-critical" by the contracting agency and alternative shade standards or information regarding shade may be specified in the contract or order (see 7.9). It is recommended that manufacturers refer to the contract or order to determine the criticality of the shade matching or alternate shade standards that are acceptable.

7.5 Sock compression testing. Compression testing follows BS 661210 Section 4 with MSC 117 modification. A known testing source is the following:

Manufacturing Solutions Center

301 Conover Station SE

Conover, NC 28613

If a supplier prefers an alternate test using BS 661210 without MSC 117 modification, the supplier should identify the circumference settings used to measure compression, which includes specifying the two circumferences used when measuring compression. The circumferences required are the ankle circumference 10 centimeters above the center gore heel and the circumference at the widest part of the calf, which requires identifying this location based upon a donned sock with proper fit.

7.6 Alternate method for cuff width. For an alternate method for cuff width, see www.legsource.com/Hosiery_Consortium_Testing/hifomaco-scissor.html.

7.7 Toxicity testing. OECD Guidelines for the Testing of Chemicals, Section 4, Test Nos. 404 and 406 are suggested to assess the requirement in 3.11, but other methods may be used.

7.8 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for a stated requirement beyond the minimum wool content and graduated compression specified in 5.1 and table V.

7.9 Ordering data. The contract or order should specify the following:

a. Title, number, and date of this commercial item description.
b. Size required (see 2.1).
c. Shade standard information (see 3.5 and 7.4.2).
d. Labeling information (see 3.9).
e. Acceptance criteria, including AQLs (see 3.10, 3.12, and 5.4).
f. When toxicity testing is required (see 3.11).
g. Regulatory requirements (see 4).
h. Product conformance provisions (see 5).
i. Packaging requirement (see 6).

7.10 Known supplier. A known supplier of the socks is the following:

Ellsworth & Company LLC, Style ECWC-03

253-301-2800

www.ellsworthsocks.com

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7.11 Use of equivalent items. Prior to using equivalent test items, fixtures, or materials, vendors should verify with the contracting agency the acceptability of the equivalent item.

7.12 Key words.

Footwear

Graduated compression

Wool

CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodians:
Army – AV
Navy – MC
Air Force – 11

Review Activities:
Army – GL, MI
Navy – AS, CG1, NU
Air Force – 03, 70
DLA – CT

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FAS

Preparing Activity:
Navy − MC
(Project 8415-2023-003)

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